

Unit I

1. Absolute Error

$$E_A = |x - x_1| = \Delta x$$

2. Relative Error

$$E_R = \frac{\text{Absolute Error}}{\text{True Value}} \quad E_R = \frac{E_A}{x} \quad E_R = \frac{\Delta x}{x}$$

3. Percentage Error

$$E_P = E_R \times 100$$

4. Bisection Method (No formula)

5. False Position

$$x_1 = \frac{a \cdot f(b) - b \cdot f(a)}{f(b) - f(a)}$$

6. Iteration Method (No formula)

7. Newton Raphson Method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

8. Finite Differences

$$\Delta y_0 = y_1 - y_0$$

$$\nabla y_1 = y_1 - y_0$$

9. Forward Difference (Newton Method)

$$y_n(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p+1)(p-2)}{3!} \Delta^3 y_0 + \dots$$

$$\Delta^3 y_0 + \dots$$

10. Backward Difference

$$y_n(x) = y_n + p \nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \dots$$

11. Interpolation Formula

$$L_1(x) = \frac{x - x_1}{x_0 - x_1} (y_0) + \frac{x - x_0}{x_1 - x_0} (y_1)$$

$$= l_0(x) y_0 + l_1(x) y_1$$

$$= \sum_{i=0}^1 l_i(x) y_i$$

12. Newton's General Interpolation

Unit II

1. Newton's Forward Differentiation

$$\left[\frac{dy}{dx} \right]_{x=x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \dots + \frac{1}{n} \Delta^n y_0 \right]$$

$$\left[\frac{d^2 y}{dx^2} \right]_{x=x_0} = \frac{1}{h^2} \left[\Delta^2 - \Delta^3 + \frac{11}{12} \Delta^4 - \frac{5}{6} \Delta^5 + \frac{137}{180} \Delta^6 - \dots \right]$$

2. Newton's Backward Differentiation

$$\left[\frac{dy}{dx} \right] = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n - \dots + \frac{1}{n} \nabla^n y_n \right]$$

$$\left[\frac{d^2 y}{dx^2} \right] = \frac{1}{h^2} \left[\nabla^2 + \nabla^3 + \frac{11}{12} \nabla^4 + \frac{5}{6} \nabla^5 - \dots \right]$$

3. Trapezoidal Rule

$$\int_{x_0}^{x_n} y \cdot dx = \frac{h}{2} \left[y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n \right]$$

$$h = \frac{b-a}{n}$$

4. Simpson's $1/3$ Rule

$$\int_{x_0}^{x_n} y dx = \frac{h}{3} \left[y_0 + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + y_n \right]$$

5. Simpson's $3/8$ Rule

$$\int_{x_0}^{x_n} f(x) dx \approx \frac{3h}{8} \left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-3}) \right]$$

Unit III

(No Formulas)

Unit IV

1. Taylor's Series

$$y(x) = y_0 + (x-x_0)y_0' + \frac{(x-x_0)^2}{2!}y_0'' + \dots$$

2. Euler's Method

3. Euler's Modified Method

$$y_1 = y_0 + \frac{h}{2} [\phi(x_0, y_0) + \phi(x_1, y_1)]$$

4. Runge-Kutta Method.

$$y_1 = y_0 + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = h\phi(x_0, y_0)$$

$$k_2 = h\phi(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_1)$$

$$k_3 = h\phi(x_0 + \frac{1}{2}h, y_0 + \frac{1}{2}k_2)$$

$$k_4 = h\phi(x_0 + h, y_0 + k_3)$$

(continued on)

VI. Find

Unit - 11

Fitting reaction

$$\sum y = na + b \sum x$$

$$\sum xy = a \sum x + b \sum x^2$$

$$\sum z = na + b \sum x + c \sum y$$

$$\sum xz = a \sum x + b \sum x^2 + c \sum xy$$

$$\sum yz = a \sum y + b \sum xy + c \sum y^2$$